

Basic Research on Ride-Hailing Analytics and Customer Churn Prediction

Prompt: I am building a ride-hailing predictive model to predict passenger churn and segmentation for focused promotion and budgeting. I have GPS (pickup/dropoff - long/lat, time, fare, tip, surge multiplier, session features, driver features, rider features, trip features

Google: To predict passenger churn and build actionable segmentations for targeted marketing, you should combine your universal RFM metrics with domain-specific features.

Below is a comprehensive framework of engineered features, categorized to directly optimize your promotional budgeting.

1. Advanced Spatio-Temporal & RFM Features (Recency & Frequency)

Standard RFM misses *where* and *why* passengers travel. Use your GPS and time data to capture behavioral patterns:

- **Recency / Frequency By Context:** Calculate separate Recency and Frequency scores for **Commute Hours** (7–9 AM, 4–6 PM) versus **Leisure/Weekend Hours**. Commuter churn indicates a loss of high-value, predictable revenue.
- **Spatial Entropy / Variety:** Count the number of unique pickup/dropoff coordinates (binned via H3/S2 geohashes). High spatial variety implies an erratic user; low variety (e.g., just Home $\rightarrow$ Work) indicates a routine user. A sudden break in a routine user's pattern is a strong churn signal.
- **Preferred Location Recency:** Days since the last trip to the user's most frequent destination (e.g., an airport or office hub).

2. Monetary, Pricing, and Price-Sensitivity Features (Monetary Spend)

These features help you identify which segments only ride when subsidized, preventing you from wasting budget on chronic discount-chasers.

- **Surge Tolerance Index:** The ratio of rides accepted under high surge (Multiplier > 1.2) to the number of times a user searched/opened the app during surge but didn't book. Low tolerance indicates a price-sensitive rider who will churn to competitors if promotions stop.
- **Tip-to-Fare Ratio:** Average tip divided by average fare. High ratios isolate affluent, low-churn segments who prioritize service quality over price.
- **Promotional Spend Elasticity:** Total historical discounts used divided by total spend. This isolates "promo-hunters" from organic, loyal users.

3. Friction & Quality of Service Features (The Primary Drivers of Churn)

Passengers rarely churn randomly; they usually leave due to poor app experiences or operational failures.

- **Session-to-Book Ratio:** Total completed rides divided by total app opens (sessions). A dropping ratio means the passenger is opening the app, seeing high ETAs or prices, and closing it to look elsewhere.
- **Cancellation Friction Rate:** The percentage of sessions where the driver cancelled on the rider, or the rider cancelled due to a long wait time.
- **Driver Rating Divergence:** The frequency of 1–3 star ratings given by the rider. A sudden bad experience is a leading indicator of immediate churn.

Feature Matrix for Segmentation & Promotional Strategy

Feature Category	High-Value Segment (Protect Budget)	Price-Sensitive Segment (Targeted Promo)	Churn-Risk Segment (Win-Back Promo)
RFM Profile	Low Recency, High Frequency, High Monetary	Moderate Recency, High Frequency during sales, Low Monetary	High Recency, Declining Frequency
Surge Multiplier	High acceptance rate during peak surge.	Low acceptance; abandons session during surge.	Highly variable; avoids surge entirely.
Spatial Behavior	Consistent routines (Home/Work hubs).	High variance or short, cheap inner-city trips.	Dropped out of their usual routine routes.
Session Ratio	High (~ 80%+ book rate per session).	Low (Opens app to compare prices across platforms).	Steep downward trend in booking ratio over last 14 days.
Action Strategy	Do not subsidise. Offer VIP perks (priority dispatch, free upgrades) to maintain loyalty.	Apply targeted promotions. Use time-bound discounts strictly during low-demand/off-peak hours.	High-value win-back. Trigger an aggressive, multi-step discount notification before they completely uninstall.

✔ Next Step Recommendation

To execute this, begin by aggregating your trip-level GPS and session data into a **User-Snapshot Table** where each row is a unique rider_id. Calculate these rolling features over 7-day, 30-day, and 90-day lookback windows to capture short-term behavioral shifts versus long-term baseline traits.

Would you like assistance writing the **SQL or Python code** to convert your raw timestamped GPS data into the spatial routine and session-to-book features?

Prompt: in data science for ride hailing analytics is recency, frequency and monetary spend domain specific engineered features, cited by sources?

Google: No, Recency, Frequency, and Monetary (RFM) metrics are not domain-specific to ride-hailing analytics.

Instead, RFM is a widely cited, generalized **customer behavior and segmentation framework**. Originating in the 1990s as a direct-mail targeting tool, it has been applied across nearly all retail, e-commerce, and service-based industries to calculate customer lifetime value and predict churn. [[1](#), [2](#), [3](#), [4](#), [5](#)]

While it is frequently used to engineer features for customer retention modeling in ride-sharing, the foundational RFM framework is completely **industry-agnostic**. [[1](#), [2](#), [3](#)]

Core RFM Concepts (Universal Framework)

- **Recency:** Time since a customer's last transaction.
- **Frequency:** Number of transactions made within a given period.
- **Monetary Value:** Total or average spend on the platform within that period. [[1](#)]

Ride-Hailing vs. Domain-Specific Features

In data science for the ride-hailing industry, relying *solely* on standard RFM metrics limits your predictive power. Practitioners typically engineer **domain-specific features** that capture the unique spatial and temporal dynamics of on-demand mobility: [[1](#)]

- **Spatio-temporal features:** Pick-up/drop-off locations, trip duration, route distance, and commute times.
- **Demand-supply metrics:** Surge pricing multipliers, wait times, and ride cancellation rates.
- **Contextual factors:** Weather conditions, local events, and time-of-day. [[1](#), [2](#), [3](#),

Academic and Industry Citations

- **Customer Behavior Theory:** Foundational RFM analysis concepts for consumer targeting and loyalty are verified and explored extensively across business literature. [[1](#), [2](#), [3](#)]
- **Mobility Industry Applications:** While standard RFM features are often used for passenger churn and customer lifetime value (CLV), literature regarding human mobility patterns and ride-hailing dynamics focuses more heavily on GPS-driven demand modeling, trip purpose inference, and order-matching strategies.

If you are planning to build a **ride-hailing predictive model**, tell me:

- Are you trying to predict **passenger churn** or **driver dispatch demand**?
- What **granularity of data** do you have available (e.g., GPS pings, ride requests, or just billing history)?

I can help identify the **most effective engineered features** for your specific use case.

